EECE 306 Lab 9 notebook. Current Sources and the Biot-Savart Law

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Current sources use the same element-list setup as the charge sources. The checks below compare the field with formulas, symmetry, and limits.

Contents

- Setup
- How long em.field.H took
- Circular loop against the closed form
- Finite segment against the infinite wire
- Solenoid, built by reusing the ring
- Reversal, linearity, and the merge policy
- Timing at N = 5000
- Interpretation
- Problems encountered
- Full test suite

Setup

```
clear; close all;
addpath(pwd);
run(fullfile('tests','lab09_studies.m'));
```

How long em.field.H took

em.field.H is the same loop as em.field.E with a different numerator.

```
minutes_for_H = 1.95;
fprintf('H review-to-first-passing-check interval: %.2f minutes\n',minutes_for_H);
% The recorded interval was about 1.9 minutes, from reviewing E through
% the first independent check of H, so it is not a timing for the whole lab.
% E already handled observation points in chunks and applied its constant
% at the end, so that part stayed the same. H reads Idl instead of q.*w,
% uses Idl cross (r-source), and divides by 4*pi instead of 4*pi*eps0.
% E itself did not need a rewrite; keeping the vector source data separate
% from the kernel made this much faster than starting over.
```

```
H review-to-first-passing-check interval: 1.95 minutes
```

Circular loop against the closed form

```
a = 1; I = 1;
ring = @(t)[a*cos(t) a*sin(t) zeros(size(t))];
sr = em.src.lineCurrent(I, ring, [0 2*pi], 400);
z = linspace(0, 2, 10)';
Hnum = em.field.H(sr, [zeros(10,2) z]);
Href = I*a^2 ./ (2*(a^2 + z.^2).^1.5);
disp('    z (m)     relative error')
fprintf('%9.4f     %.3e\n',[z abs(Hnum(:,3)-Href)./Href]');
Hc = em.field.H(sr, [0 0 0]);
fprintf('center H_z = %.6f   I over 2a = %.6f\n', Hc(3), I/(2*a));
fprintf('transverse component at center = %.3e  (consistent with zero)\n', hypot(Hc(1), Hc(2)));
```

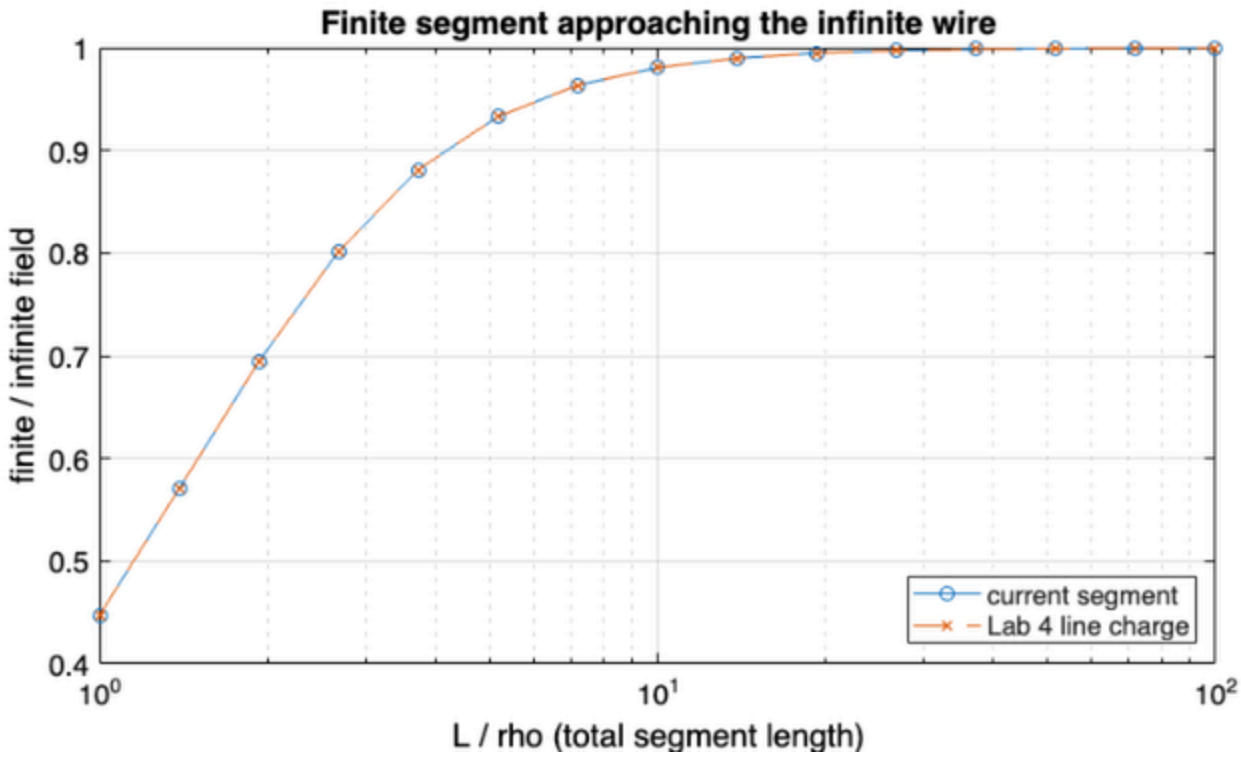
```
     z (m)      relative error
    0.0000      1.049e-11
    0.2222      1.049e-11
    0.4444      1.049e-11
    0.6667      1.049e-11
    0.8889      1.049e-11
    1.1111      1.049e-11
    1.3333      1.049e-11
    1.5556      1.049e-11
    1.7778      1.049e-11
    2.0000      1.049e-11
center H_z = 0.500000   I over 2a = 0.500000
transverse component at center = 0.000e+00  (consistent with zero)
```

Finite segment against the infinite wire

The same study as the Lab 4 finite line, with the field engine swapped. The curve has the same shape, and the handout asks why.

```
rho = 1;
hor = S9.Lrho; ratio = S9.wireRatio;
figure; semilogx(hor,ratio,'o-',hor,S9.chargeRatio,'x--'); grid on
xlabel('L / rho (total segment length)'); ylabel('finite / infinite field')
legend('current segment','Lab 4 line charge','Location','southeast')
title('Finite segment approaching the infinite wire')
fprintf('first sampled 99%%: L/rho %.3f; half-length h/rho %.3f\n', ...
    S9.wireThreshold,S9.wireThreshold/2);
fprintf('continuous threshold: L/rho %.3f; half-length h/rho %.3f\n', ...
    S9.wireCritical,S9.wireCritical/2);
% The magnetic and Lab 4 electric curves overlap because their transverse
% kernels have the same distance dependence. The starter called the full
% length h, while the handout uses 2h; here L is the full length, so the
% thresholds match only after that factor of two is accounted for.
```

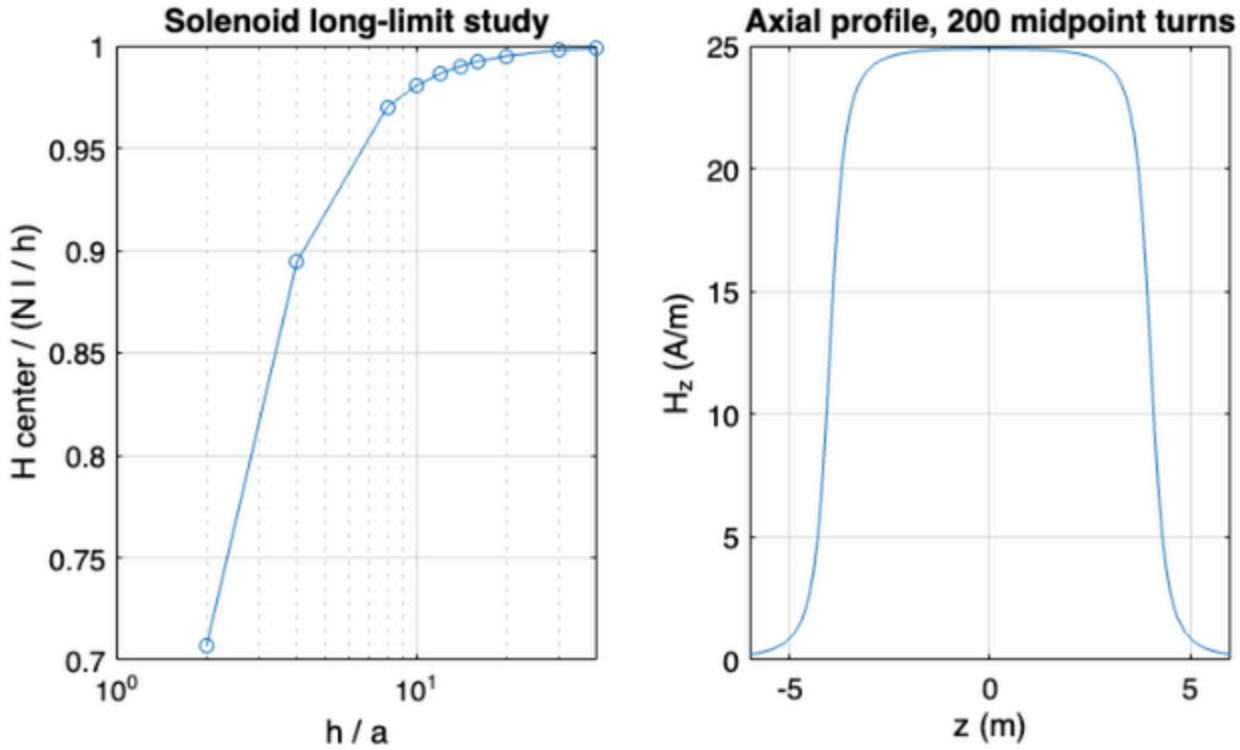
```
first sampled 99%: L/rho 19.307; half-length h/rho 9.653
continuous threshold: L/rho 14.036; half-length h/rho 7.018
```



Solenoid, built by reusing the ring

```
aS = S9.a; hS = S9.height; Nturns = S9.turns;
sol = S9.sol; zax = S9.zax; Hax = S9.Hax;
figure;
subplot(1,2,1); semilogx(S9.aspect,S9.solRatio,'o-'); grid on
xlabel('h / a'); ylabel('H center / (N I / h)')
title('Solenoid long-limit study')
subplot(1,2,2); plot(zax,Hax(:,3),'-'); grid on
xlabel('z (m)'); ylabel('H_z (A/m)')
title('Axial profile, 200 midpoint turns')
fprintf('solenoid first sampled 99%%: h/a %.1f\n',S9.solThreshold);
fprintf('center H_z %.5f; NI/h %.5f; ratio %.5f\n', ...
    S9.center,Nturns*I/hS,S9.center/(Nturns*I/hS));
fprintf('mouth / center %.5f; turn-refinement change %.3e\n', ...
    S9.mouthRatio,S9.turnRefinement);
% We used 200 evenly spaced midpoint turns instead of 25 endpoint turns.
% That keeps the turn density close to N/h and resolves the winding, so
% the 99% crossing is a long-solenoid limit rather than a spacing artifact.
% The mouth field is about half the center field, and it drops outside.
```

```
solenoid first sampled 99%: h/a 16.0
center H_z 24.87593; NI/h 25.00000; ratio 0.99504
mouth / center 0.50187; turn-refinement change 9.189e-08
```



Reversal, linearity, and the merge policy

```
sneg = em.src.lineCurrent(-I, ring, [0 2*pi], 400);
fprintf('max |H(-I) + H(I)|  = %.3e  (reversal)\n', ...
    max(em.vec.mag(em.field.H(sneg, [0.3 0.2 0.5]) + em.field.H(sr, [0.3 0.2 0.5]))));
s2I = em.src.lineCurrent(2*I, ring, [0 2*pi], 400);
fprintf('max |H(2I) - 2H(I)| = %.3e  (linearity)\n', ...
    max(em.vec.mag(em.field.H(s2I, [0.3 0.2 0.5]) - 2*em.field.H(sr, [0.3 0.2 0.5]))));
sm = em.src.merge(sr, s2I);
fprintf('merged current source element count = %d  (sum of parts = %d)\n', ...
    size(sm.pos,1), size(sr.pos,1) + size(s2I.pos,1));
try
    em.src.merge(sr, em.src.pointCharge(1e-9, [0 0 0]));
    disp('ERROR, this line must not be reached')
```

```
catch err
    fprintf('merging current with charge correctly threw. %s\n', err.message);
end
```

```
max |H(-I) + H(I)|  = 0.000e+00  (reversal)
max |H(2I) - 2H(I)| = 0.000e+00  (linearity)
merged current source element count = 800  (sum of parts = 800)
merging current with charge correctly threw. Cannot merge source s2 of type 'charge' with type 'current'.
```

Timing at N = 5000

```
pts = [5+sin((1:5000)'*.1), 5+cos((1:5000)'*.2), ...
    5+sin((1:5000)'*.3)];
tic; em.field.H(sr, pts); t = toc;
fprintf('em.field.H on 5000 points took %.3f s  (requirement, under 2 s)\n', t);
```

```
em.field.H on 5000 points took 0.063 s  (requirement, under 2 s)
```

Interpretation

The source list kept the geometry out of the field engine, and the quadrature code already knew how to turn a curve into weighted elements. We added an optional tangent output to quad.line without changing its existing positions or length weights, so lineCurrent reuses that work. The surface-current source uses K times area, while the line source keeps the tangent direction. For the next interface, we would keep the directed element data available instead of reducing everything to a magnitude.

Problems encountered

The full-length versus half-length convention could have made the wire threshold look wrong by a factor of two. The first sampled crossing is only a bracket; it is not the exact continuous 99% aspect ratio. The original sparse endpoint-turn stack also changes the turn density, so we used midpoint turns and checked them against a finer winding. Near-zero transverse components need an absolute tolerance: a relative error there mostly measures division by cancellation noise. Reversing a current flips the arrows; B is mu0*mur times H, not H itself.

Full test suite

```
runTests
```

```
EECE 306 toolbox test suite
------------------------------------------------------------

eta =

  376.7303

  test_lab01 checks complete
  PASS  test_lab01            0.55 s
  test_lab02 checks complete
  PASS  test_lab02            0.12 s
  5000-point field evaluation: 0.0013 s
  test_lab03 checks complete
  PASS  test_lab03            0.04 s
  test_lab04 checks complete
  PASS  test_lab04            0.24 s
  merged source: flux 5.00000000e-09 C, enclosed 5.00000000e-09 C
  density errors at 128000 / 1024000 elements: 0.953% / 1.715% max
  sphere / cube / closed hemisphere: 3.00000000e-09 / 3.00000000e-09 / 3.00000000e-09 C
  sphere Gauss rel err 1.42e-11; midpoint N=960 rel err 4.46e-07
  test_lab05 checks complete
  PASS  test_lab05            1.46 s
  Lab 6: max absolute map error 7.776e-04 V/m
  max closed residual / (max|E| L) 2.473e-13
  Square vs exact-tangent edges 4.178e-11 V
  open midpoint errors [0.0004329 0.0001082 2.706e-05 6.764e-06] V (order 2)
  Fixed line refinement 4.965e-06 V
```

```
  growing line potentials 25.9495 -> 37.6527 V
  warning verified
  Step minimum h=3.162e-06 m, error 1.025e-11
  default error 3.586e-11
  10000 points V/grad 0.001/0.018 s
  test_lab06 checks complete
  PASS  test_lab06            1.61 s
  Lab 7: MMS error 8.224e-05, order 2.042
  Jacobi / GS / SOR iterations 1320 / 689 / 77
  Gauss flux 2.5947e-09 C, enclosed 2.5000e-09 C
  test_lab07 checks complete
  PASS  test_lab07            0.32 s
  Lab 8: plate error 1.110e-16; constant-layer error 4.441e-16
  Graded-layer orders harmonic 2.000; arithmetic 0.996
  Coax finest error 1.373%; order 0.859
  Voltage ratio 1.0000000000; filled ratio 3.0000000000
  Bar R 0.01000000 ohm m; tangential E jump 3.109e-14 V/m
  test_lab08 checks complete
  PASS  test_lab08            1.25 s
  Lab 9: loop max relative error 1.049e-11; transverse 4.901e-15 A/m
  Wire first 99%: L/rho 19.307 (half-length h/rho 9.653)
  Solenoid first 99%: h/a 16.0; mouth/center 0.501867
  H timing: 5000 points 0.035 s; 10000 points 0.066 s
  test_lab09 checks complete
  PASS  test_lab09            0.49 s
------------------------------------------------------------------
  9 passed, 0 failed, 9 total
  All tests passing.
```

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